

How Fragile Is Your Traffic-Sign Classifier?

Digital and physical adversarial attacks on an ADAS perception module

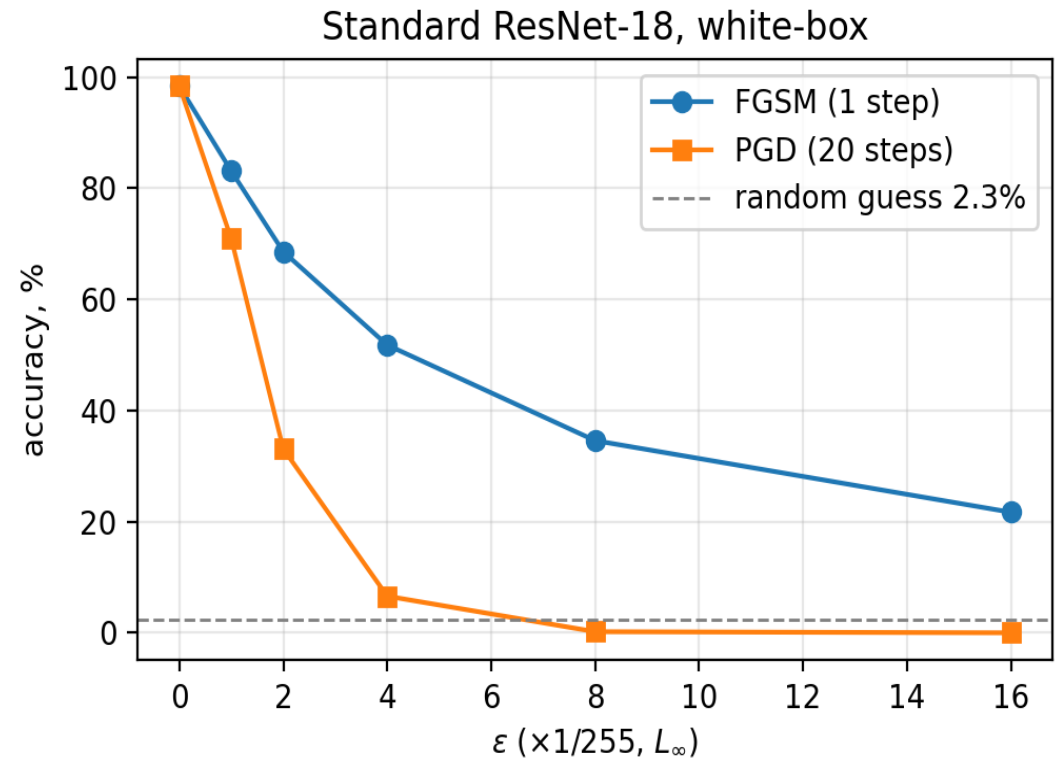
Manish · Direction: Computer Vision / NN Robustness · GTSRB, 43 classes

97.8% clean accuracy → 0.3% under PGD → 71% attack success with a printable sticker

tl;dr

motivation · main idea · results

- **Motivation: a misread STOP sign is a safety-critical failure in an L2–L3 ADAS stack.**
- Main idea: measure accuracy under an adversary, not just on a clean test set.
- v1 ResNet-18: 98.28% clean — looks shippable.
- v2 PGD-20 at $\epsilon = 8/255$: accuracy collapses to 0.2% (random guess = 2.3%).
- v3 Universal printable patch covering 4.1% of the sign: 36.91% targeted success on STOP signs.
- v4 PGD adversarial training: PGD accuracy 0.2% $\rightarrow$ 51.3%, clean 98.28% $\rightarrow$ 82.82%.
- **Conclusion: robustness is a purchase — paid in clean accuracy and $\sim 8\times$ training compute.**



Task statement & why it is reasonable

rubric: clear AI task, motivation, benefits

▪ **Task: 43-class German traffic-sign classification (GTSRB), 64×64 RGB input.**▪ Role in the Autonomous Driving stack: sensing → perception → localisation → prediction → planning → control. Sign recognition feeds speed-limit assist and intersection behaviour, so it directly changes the ego trajectory.▪ SAE automation levels: at L2 the human is the fallback; from L3 upward the machine is accountable within its ODD, so a perception failure stops being an inconvenience and becomes a liability.▪ **Threat model: the attacker has no access to weights at inference time and no access to the camera buffer — only the ability to place a physical object in the scene.** Cost: one printed sticker.▪ **Benefit of the study: turns "is it safe?" into measurable quantities — robust accuracy vs ϵ , targeted attack success rate vs patch area, and the explicit price of the defense.**

Why the trending solution over the classical one

rubric: classical baseline + objective metrics

Approach	Features	Clean Accuracy
v0 — HOG + linear SVM (2005-era pipeline)	Hand-crafted gradient histograms	87.07%
v1 — ResNet-18, ImageNet-pretrained, fine-tuned	Learned hierarchical features	98.28%

- Gap of 11.21 accuracy points — nobody would ship the classical model.
- Deep learning wins because the sign manifold (illumination, motion blur, occlusion, weather, viewing angle) is not well described by a fixed descriptor. A CNN learns translation-equivariant features end-to-end; HOG cannot adapt its filters to the data.
- **But: the deep model is the one that turns out to be catastrophically attackable. The accuracy ranking and the robustness ranking disagree — that is the point of this project.**

Literature review

the six papers this project reproduces

Work	Contribution	Used Here As
Szegedy et al. 2013 — Intriguing properties	Adversarial examples exist for NNs	Problem statement
Goodfellow et al. 2014 — FGSM	One-step linear-approximation attack	v2 weak baseline attack
Madry et al. 2017 — PGD / min-max	Strong iterative attack + adversarial training	v2 attack, v4 defense
Brown et al. 2017 — Adversarial Patch	Universal, localised, image-agnostic patch	v3 attack form
Athalye et al. 2018 — EOT	Optimise over a distribution of transforms	v3 physical realisability
Eykholt et al. 2018 — RP2	Printed stickers fool real sign classifiers	Threat model & motivation

Tooling: PyTorch · torchvision ResNet-18 · torchattacks 3.5.1 (FGSM/PGD) · scikit-image HOG · scikit-learn LinearSVC

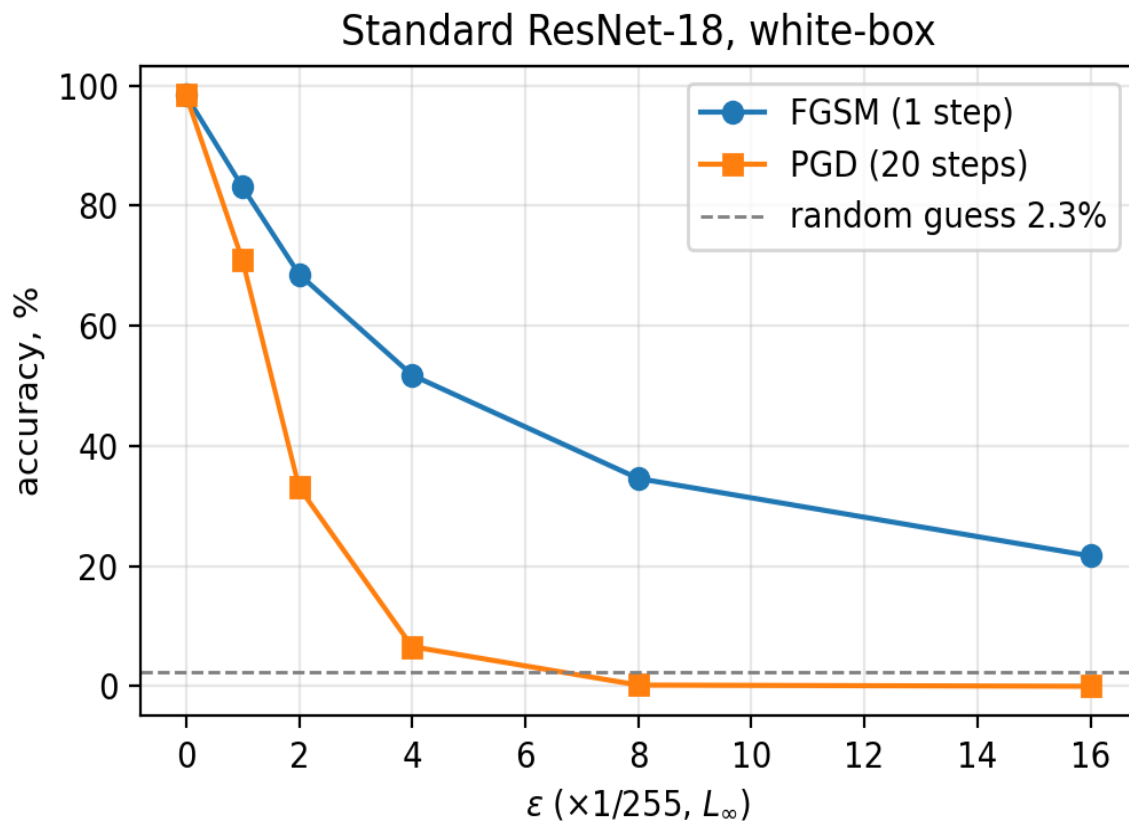
Approach & experimental setup

reproducible: fixed seed, held-out splits, normalization inside the model

▪ **Data: GTSRB, 26,640 train / 12,630 test, resized to 64×64, augmented with random affine + colour jitter.** ▪ Normalization is a layer inside the model, so every attack operates in true $[0,1]$ pixel space. This is the standard pitfall — attacking normalized space silently changes the effective ϵ budget and inflates reported robustness. ▪ Digital threat model: white-box L^∞ ball, $\epsilon \in \{1, 2, 4, 8, 16\}/255$. FGSM (1 step) and PGD (20 steps, random start, $\alpha = \epsilon/4$). ▪ Physical threat model: one universal, targeted patch (STOP $\rightarrow$ "80 km/h"), sigmoid-parameterised, optimised with Adam, composited onto the image by a differentiable affine warp so gradients flow to the patch pixels. ▪ EOT: every optimisation step samples scale 0.75–1.30, rotation $\pm 25^\circ$, translation, brightness 0.65–1.25 and additive bias — so the gradient averages over a distribution of physical conditions instead of one ideal placement. ▪ **Metrics: clean accuracy · robust accuracy vs ϵ · targeted Attack Success Rate (ASR) · ASR vs patch area. All attacks on a fixed 1,000-image subset for comparability; seed 0 throughout.**

v2 — Digital white-box attacks

FGSM vs PGD-20, L^∞



$\epsilon (\times 1/255)$	0	1	2	4	8	16
FGSM %	98.4	83.1	68.6	51.8	34.6	21.7
PGD-20 %	98.4	70.9	33.1	6.6	0.2	0.0

- At $\epsilon=8/255$ (invisible to a human): 34.6% under FGSM, 0.2% under PGD-20. Below random guessing (2.3%).
- PGD dominates FGSM at every ϵ . Had I evaluated with FGSM only, I would have claimed 34.6% "robustness" that does not exist.
- **Methodological lesson: always report the strongest attack you can afford. Weak-attack evaluation is how defenses get published and then broken.**

v2b — Is the classical model safer?

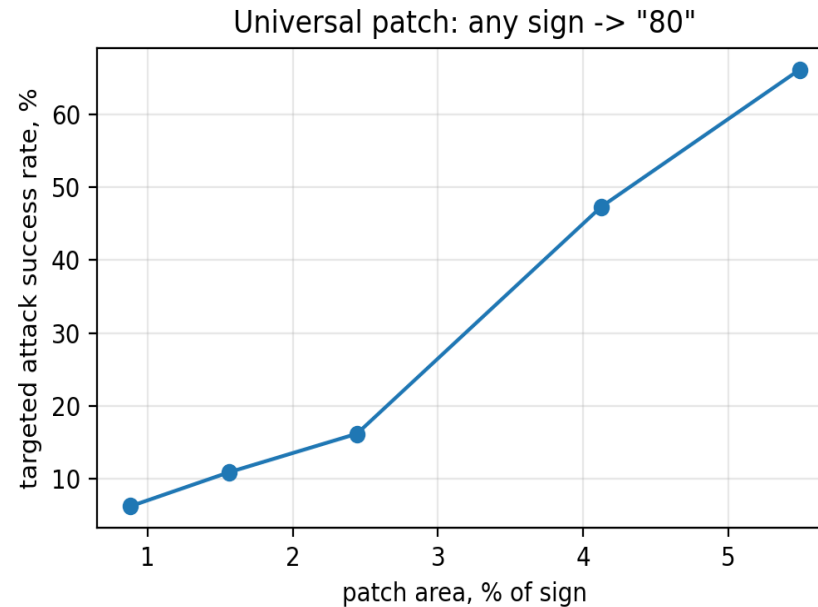
transfer the CNN-crafted PGD examples to HOG + SVM

Model	Clean (same 1,000 imgs)	On CNN-crafted PGD $\epsilon=8/255$
ResNet-18 (the attacked model, white-box)	98.4%	0.2%
HOG + linear SVM (never attacked, black-box)	86.0%	56.6%

- Fragility is largely model-specific: the perturbation was optimised against ResNet-18 gradients, and a completely different feature extractor keeps most of its accuracy.▪ But transfer is real, not zero: the SVM loses 29.4 points without the attacker ever touching it. Adversarial directions are partly shared across models trained on the same data.▪ **So "use a different or secret model" is obfuscation, not a defense — and it costs 12.4 points of clean accuracy up front. This is exactly why the next step moves to a black-box, physically realisable attack.**

v3 — Universal physical patch with EOT

one printable sticker, targeted: any sign $\rightarrow$ "80 km/h"



Patch base (px)	6	8	10	13	15
Area (% of sign)	0.9	1.6	2.4	4.1	5.5
Targeted ASR (%)	6.3	10.9	16.2	47.3	66.1

▪ **Headline:** 13×13 px patch = 4.1% of sign area $\rightarrow$ 36.91% targeted ASR on STOP signs, 51.02% across all 43 classes.▪ The patch is universal (one image, not per-input), targeted (attacker chooses output class), and localised (no camera access — just a printer).▪ ASR grows monotonically with area: an explicit attacker cost curve, not a yes/no result.▪ **EOT ablation — the digital-to-physical gap:** — Patch trained WITHOUT EOT: 80.42% ASR under ideal centred placement... — ...but only 38.46% once it is rotated, rescaled and relit.▪ **A perturbation that ignores the physical channel loses roughly half its power. EOT is what makes an attack survive a camera.**

Same signs, three input conditions

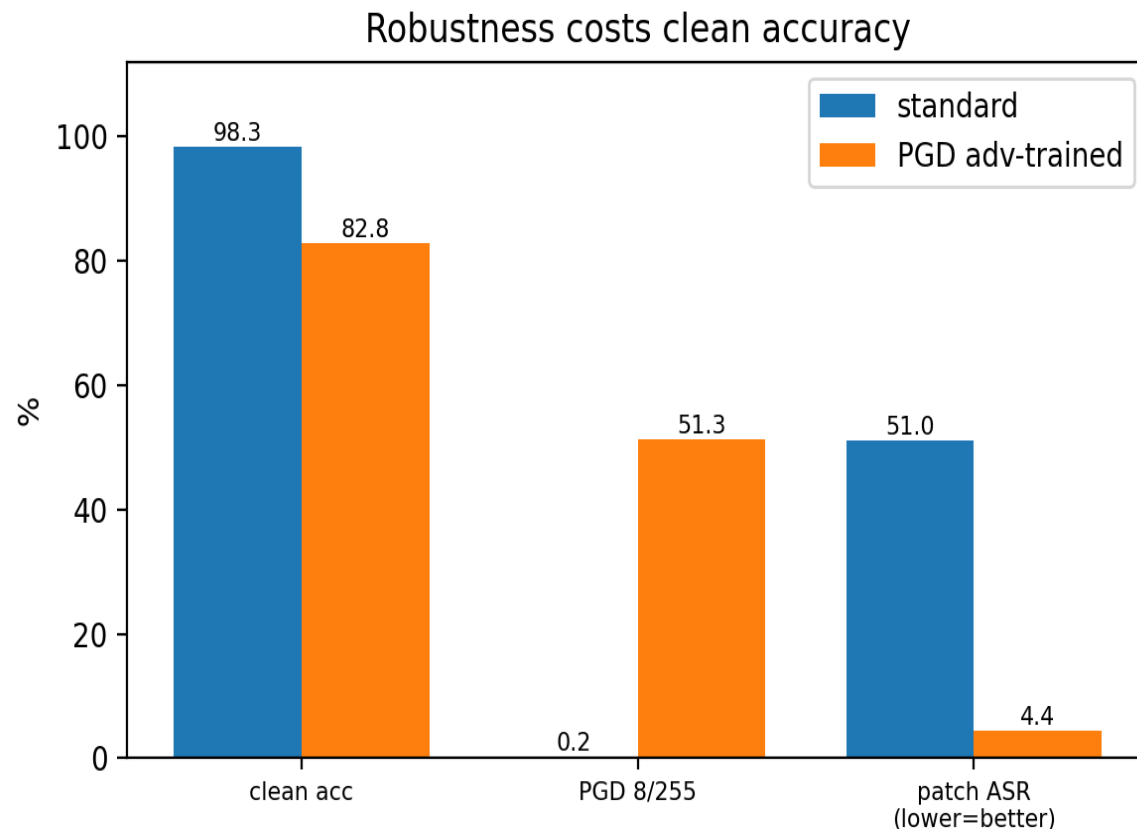
green = correct, red = misclassified



- Row 1 clean $\rightarrow$ correct. Row 2 PGD $\epsilon=8/255 \rightarrow$ wrong, and the perturbation is barely visible. Row 3 patch under EOT $\rightarrow$ driven to the attacker-chosen target class.

v4 — PGD adversarial training, and what it costs

min-max training (Madry et al. 2017)



Model	Clean %	PGD 8/255 %	Patch ASR %
ResNet-18 standard	98.28	0.2	51.02
ResNet-18 PGD adv-trained	82.82	51.3	4.39

- Objective: min over θ of max over $\|\delta\|_\infty \leq \epsilon$ of the loss. Inner maximisation = 7-step PGD per batch $\rightarrow$ roughly 8 $\times$ the training FLOPs of standard training.
- Robust accuracy rises by tens of points; clean accuracy falls by a few. This is the documented robustness–accuracy trade-off, reproduced on my own data.
- The patch attack is reduced but not eliminated: L_∞ adversarial training does not certify against a large-norm, localised perturbation. The threat models do not match.
- **Engineering conclusion: robustness is bought with clean accuracy, compute, and a threat model you must choose in advance. There is no free defense.**

Iterative process — each version changed the threat model

rubric: dealing with emerging paradigms / change in trends

Version	What I Built	Result	Why I Moved On
v0	HOG + linear SVM	87.07% clean	Not competitive → justifies deep learning
v1	ResNet-18 fine-tune	98.28% clean	Looks shippable — so I tried to break it
v2	FGSM / PGD, L^∞	PGD@8/255 → 0.2%	White-box pixel access is unrealistic threat
v2b	Transfer PGD → SVM	86.0% → 56.6%	Obscurity is not a defense; need black-box attack
v3	Universal patch + EOT	36.91% ASR on STOP	Physically realisable → now needs a defense
v4	PGD adversarial training	PGD@8/255 → 51.3%	Works, but costs accuracy + 8× compute

- The paradigm shift here is not in the architecture — it is in the threat model. Every redesign was driven by asking "what can the attacker actually do?"

Conclusions, limitations, next steps

1. Clean accuracy is not a safety argument: 98.28% → 0.2% under an imperceptible L^∞ perturbation.
2. Always evaluate with the strongest attack: FGSM alone would have overstated robustness by 34 points.
3. Attacks are physically realisable: a 4.1%-area printed patch reaches 36.91% targeted ASR on STOP signs, and EOT is what buys transfer across the physical channel (80.42% → 38.46% without it).
4. Defenses are purchases, and they are threat-model-specific: L^∞ adversarial training does not fully cover a localised patch.▪ *Limitations: classifier only, no detection stage; 64×64 inputs; single architecture family; the physical demo is a small qualitative sample, not a controlled distance/angle study; adversarial training used a reduced PGD-step budget for compute reasons.*▪ *Next steps: attack a detector end-to-end (RP2-style); CNN vs ViT robustness comparison (different inductive bias, different attack surface); randomised smoothing for certified radii; patch-aware defenses (Local Gradient Smoothing, PatchGuard).*

Code, artifacts & references

▪ **Code + slides + generated figures:** Notebook: notebooks/01_full_pipeline.ipynb — runs top-to-bottom on a Colab T4 GPU, fixed seed.▪

Artifacts: results/summary.csv, results/digital_attacks.csv, results/asr_vs_area.csv, results/results.json, figs/*.png, physical/print_me_patch.png, cnn_std.pt, cnn_adv.pt▪

Dataset: GTSRB — Stallkamp et al., 2012, via torchvision.datasets.GTSRB▪

Libraries: PyTorch · torchvision · torchattacks 3.5.1 · scikit-image · scikit-learn▪

Papers: Szegedy 2013 (arXiv 1312.6199) ·

Goodfellow 2014 (1412.6572) ·

Madry 2017 (1706.06083) ·

Brown 2017 (1712.09665) ·

Athalye 2018 (1707.07397) ·

Eykholt 2018 (1707.08945)

Link to code - <https://github.com/manish0338/ai-hw-Summer-2026-nn/blob/main/mlp.ipynb>

Link to artifacts - <https://github.com/manish0338/ai-hw-Summer-2026-nn/tree/main/artifacts>

Appendix A — Full digital attack data

standard + adversarially-trained models

ϵ ($\times 1/255$)	Std FGSM	Std PGD-20	Adv FGSM	Adv PGD-20
0	98.4	98.4	83.6	83.6
1	83.1	70.9	79.7	79.6
2	68.6	33.1	78.0	77.3
4	51.8	6.6	71.5	70.1
8	34.6	0.2	60.5	51.3
16	21.7	0.0	42.8	20.4

- All evaluations on a fixed 1,000-image subset (seed 0) for comparability.
- PGD: 20 steps, random start, $\alpha = \epsilon/4$. FGSM: 1 step.
- Adversarially-trained model uses warm-start from cnn_std.pt + 3–7 step PGD inner loop.

Appendix B — Patch attack details & EOT ablation

universal targeted patch: any sign → "80 km/h" (class 5)

Patch base (px)	Area (% of sign)	Targeted ASR (%)
6	0.9	6.3
8	1.6	10.9
10	2.4	16.2
13	4.1	47.3
15	5.5	66.1

EOT Ablation

Condition	ASR (%)
EOT patch, all 43 classes	51.02
EOT patch, STOP signs only	36.91
No-EOT patch, ideal centred placement	80.42
No-EOT patch, under random transforms	38.46
Adv-trained model, EOT patch	4.39

▪ Patch trained with Adam (lr=0.05), 300–400 iterations, sigmoid parameterisation.▪ EOT samples: scale 0.75–1.30, rotation $\pm 25^\circ$, translation [-0.40, 0.40] / [-0.05, 0.45], brightness 0.65–1.25, additive bias ± 0.08 .▪ **The 80.42% → 38.46% drop when applying transforms to a non-EOT patch quantifies the digital-to-physical gap.**▪ All ASR measurements on fixed 1,000-image eval subset, 2–3 repetitions averaged.